one_round_SDUA | SDUA | output | retain | 756 input tokens, 146 scored

<|start|>system<|message|> systems tend to inflict impact damage on particles as they encounter bends in a pipeline. The resulting effects from either types of damage are dictated by the bulk properties and can range from 'smearing' and adhesion (ultimately leading jamming or blockage) or an increase in fines content within a size distribution which can contribute to flow problems and increased segregation effects.

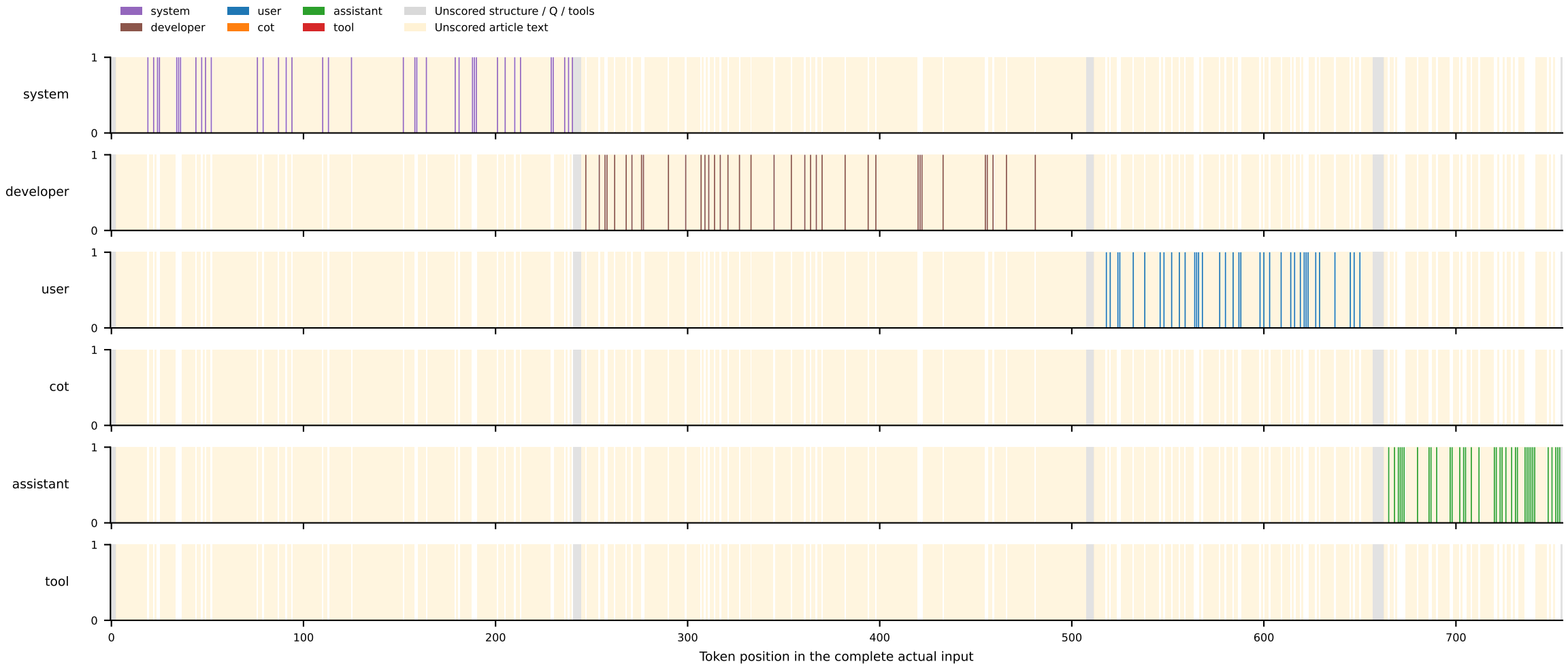
Some materials are very sensitive to length of time that they are left in storage and can form lumps of solid material ranging in size from millimetres up to complete big bags! Similarly lump strength can range from weak enough to disassemble by hand pressure up to strength comparable to set concrete in some cases. The mechanisms behind lump formation are varied and it is not unusual for different mechanisms to act together. Sometimes useful improvement can be obtained by controlling the temperature of the bulk solid as it leaves the process and on other occasions a resolution exists in the redesign of storage equipment with the objective of either controlling the stresses acting on the stored bulk solid or, alternatively, using the storage vessel as means to 'condition' the bulk solid by effecting a slow removal of excess moisture (where this is primary driving force behind the problem). <|end|><|start|>developer<|message|> 'smearing' and adhesion (ultimately leading jamming or blockage) or an increase in fines content within a size distribution which can contribute to flow problems and increased segregation effects.

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Where wide variations in size, density or shape occur within a bulk solid it is highly likely that a reduction in homogeneity will develop through handling and storage steps within a process. The mode of delivering material into a reception silo (pneumatic or mechanical) and the location at which material enters the<|end|><|start|>user<|message|> in the redesign of storage equipment with the objective of either controlling the stresses acting on the stored bulk solid or, alternatively, using the storage vessel as means to 'condition' the bulk solid by effecting a slow removal of excess moisture (where this is primary driving force behind the problem).

Where wide variations in size, density or shape occur within a bulk solid it is highly likely that a reduction in homogeneity will develop through handling and storage steps within a process. The mode of delivering material into a reception silo (pneumatic or mechanical) and the location at which material enters the reception vessel/silo will have strong influence over the severity of segregation that will occur as the material is deposited. In concert with<|end|><|start|>assistant<|channel|>final<|message|>lict impact damage on particles as they encounter bends in a pipeline. The resulting effects from either types of damage are dictated by the bulk properties and can range from 'smearing' and adhesion (ultimately leading jamming or blockage) or an increase in fines content within a size distribution which can contribute to flow problems and increased segregation effects.

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Native Harmony; colored strokes = scored article tokens. Gray = structure; pale yellow = unscored article text.